Task 2: Careers and Career Goals

For Task 2 of D596: The Data Analytics Journey, I will create a career plan, compare three different data analytics disciplines as described by ProjectPro, and identify a potential career goal in my career plan based on my strengths and academic/MSDA track (Data Engineering) interest.

Career Plan

A. **Create a career plan.**

My short-term career plan is to change from a business analyst to a data analyst within my current or a new organization upon completing the MSDA program. A data analyst's responsibilities and daily tasks align more closely with my interests than a business analyst's. These responsibilities will offer me more engaging and fulfilling work throughout my career. Based on the U.S. Bureau of Labor Statistics (BLS), employment growth in data science and related fields is projected to grow by 36% through 2033, outpacing the 11% growth projected for management analysts. (U.S. BLS, 2025).

While pursuing a master's in data analytics, I aim to improve my skills with data analysis tools like Python, SQL, and Tableau. I also plan to develop new competencies with other data analyst tools and techniques, like using R for analysis and performing predictive analysis.

In the long term, my career plan is to advance into data engineering; in this role, I will build and maintain systems that process, store, and retrieve data. The MSDA program's data engineering specialization will prepare me for this role by improving my working knowledge in data architecture, data pipelines and integration, and cloud computing. Pursuing the MSDA demonstrates to future employers my dedication to the data analytics and engineering career.

A1. Describe three different roles or careers in data analytics.

The data analytics umbrella covers a wide variety of roles across many industries. The three roles that stick out to me the most would be the **Data Analyst**, **Data Engineer**, and **Data Scientist**.

1) Data Analyst:

a) When beginning a new project, the data analyst typically follows the data analytics life cycle (DALC). After the business understanding phase, the data analyst's responsibilities are to collect, clean, prepare, and process data for analysis. They will use SQL, Python, Excel, or Tableau to explore data and present their findings. Data analysts may also use techniques like predictive analysis, which involves algorithms to assist the data analyst in their predictions. Data analysts may also create dashboards and reports to provide insight to stakeholders so they can make data-driven decisions for the organization.

2) **Data Engineer:**

   a) Data engineers are responsible for designing, building, and maintaining an organization's data processing, storage, and retrieval systems – data warehouses and pipelines. These duties require data engineers to collaborate with other data specialists like data scientists and analysts. Data engineers will work with these different data specialists to ensure that the infrastructure meets the current and future analytical needs. Data engineers will deploy various tools to achieve their goals, such as Apache Spark, SQL, and cloud platforms like AWS or Azure, to handle large data operations and ensure data quality and security.

3) **Data Scientist:**

   a) Data scientists work to uncover patterns and trends by collecting and processing large and complex datasets so that they may obtain meaningful insight and create predictive models. Before analyzing a data set, the scientist will clean and normalize the data. Often going further than a data analyst might by applying advanced statistical methods, machine learning algorithms, and utilizing Python or R programming at a high level. Typical responsibilities for data scientists include developing predictive models, charts, graphs, and dashboards to share their insights with stakeholders. They also build forecast models, perform A/B tests, or develop automation solutions for business needs – for example, helpdesk chatbots or fraud detection systems.

a) Discuss the differences between the roles or careers from part A1

The three roles selected in part A1, **data analysts**, **data scientist**, and **data engineer**, can have overlapping and shared responsibilities within some organizations. However, each role maintains its duties within an organization.

A **data analyst** seeks to answer 'why' or 'how' events occur and identify the correlation within historical data to create predictions on future events. They often build models and future forecasts to assist in guiding business decisions.

For example, a data analyst for a car manufacturer might discover that sports car sales increase during the spring and summer months. With this insight, the data analyst could guide the company in prioritizing the distribution of specific models during peak seasons to maximize sales and reduce customer waiting times.

A **data scientist** works with more complex data sets, both structured and unstructured, to build advanced statistical and machine learning models to find hidden trends and develop predictive tools. Data scientists seek to explain past events, anticipate future outcomes, and provide stakeholders with strategic recommendations based on their gained insight.

A **data engineer** collaborates with other teams within organizations to understand the business needs so they may design, implement, and maintain appropriate data collection, processing, and storage systems that are both accessible and secure. Engineers typically support multiple data storage systems, such as data lakes and warehouses, to accommodate the organization's data needs for raw mass data storage (lake) and structured (warehouse) data ready for analysis. With data pipelines, engineers design an Extract, transform, and load process to transform the raw data into a structured dataset that analysts and scientists can utilize more easily than raw data.

A2. Describe how each role from part A1 supports the data analytics life cycle.

1. **Data analysts** support multiple phases of the data analytics life cycle (DALC). During the discovery phase, they meet with stakeholders to define business goals, answer key questions, and identify measurable metrics. In the data collection phase, data analysts will gather datasets from various sources utilizing tools such as SQL, web scraping, and surveying. Afterward, data analysts will work through the data cleaning phase before performing data analysis (data exploration). Data analysts can create predictions with the use of predictive modeling so they may gain insight into future trends to support strategic decision-making for stakeholders.
2. **Data Scientists** can begin their work with unstructured or structured data and are involved in the data cleaning and exploration phases. They use Python or R programming languages to clean and prepare data for advanced analysis. Data scientists conduct deep dives into datasets during the exploration phase to discover complex relationships and trends. They may also contribute to the modeling phase by building machine-learning models that provide stakeholders with actionable insight.
3. **Data Engineers** provide a technical foundation for the DALC by planning, building, and maintaining the organization's data systems. In the discovery phase, data engineers collaborate with data analysts and data scientists to understand the needs of the business and identify relevant data sources. Data Engineers build and maintain the data pipeline ETL/ELT (Extract, Transform, Load) processes to ingest large raw data sets and transform them into clean, usable datasets. Data Engineers must maintain data lineage for regulatory and audit reporting requirements across many industries

ProjectPro Data Analytics Disciplines

B. Compare three different data analytics disciplines as described by ProjectPro.

I will compare three data analytics disciplines from ProjectPro: **Business Analytics**, **Data analysis**, and **Data Engineering**.

ProjectPro describes the three disciplines of Business Analytics, Data Analysis, and Data Engineering well and how each serves a distinct organizational function.

1. **Business analysts** seek to identify problems and possible blockers and provide the business with solutions. A business analyst's work is typically light on coding and relies more heavily on business intelligence and some statistical analysis. Business analysts do not often find it necessary to interact with raw datasets and perform their analysis work on structured data.
2. **Data analysts** seek to identify correlations between data and form predictions on the future of certain events, allowing the business to prepare or react appropriately.
3. **Data engineers** are responsible for the business's data architecture, which includes the business databases and processing systems. Data engineers are responsible for securing these architectures from outside and unauthorized users while permitting access by data professionals within the organization. Data engineer processing systems handle raw data at different scales. They will convert this data into a formatted, structured data set that data analysts and scientists can use for further analysis.

B1. Identify three types of careers from the Bureau of Labor and Statistics government data in your career plan.

Three careers from the Bureau of Labor and Statistics in my career plan in data analytics are **economist**, **financial analyst**, and **software developer**. (U.S. BLS, 2025).

B2. Identify your academic skills and needs for the careers considered in part B1.

a. Economist

1. **Academic Skills**
    1. While some may find job opportunities as economists with bachelor's degrees, it is common for a master's degree to be required to be an economist or even a Ph.D. in some roles. Economists must be proficient in higher-level mathematics, such as calculus and linear algebra, as they will use these to develop analysis models. Economists also need solid computer skills to work with software used in data analysis.
1. **Needs**
    1. I am currently on the path to earning my master's degree, which will satisfy the requirement for a higher-level degree as an economist. The MSDA program will also prepare me to use software and write scripts to perform data analysis at a higher level and incorporate different statistical methods as I develop models.

b. Financial Analyst

1. **Academic Skills**
    1. The entry-level financial analyst role welcomes candidates who possess a bachelor's degree. However, some employers may require a master's degree. Specific financial analyst roles may require certain industry certifications, such as the Chartered Financial Analyst (CFA) cert. However, these certifications are rarely needed for entry-level roles, and the employer typically covers the associated certification cost. Financial analysts must be proficient with computers and able to use basic software to analyze financial trends. Financial analysts also need strong

math skills since they often have to estimate the value of different financial securities using advanced calculations.

2. **Needs**

    1. I currently meet the degree requirements from an entry-level perspective. However, the MSDA degree will improve the chances of selection for an interview over other candidates. The statistical methods used in the MSDA program will strengthen my mathematical skills and thinking. With this, in combination with the coding and scripting requirements of the MSDA, I will gain a beneficial perspective on using different software to perform analysis.

c. Software Developer

1. **Academic Skills**

    1. A master's degree is rarely required to become a software developer. Most employers see a bachelor's degree as adequate. However, a master's may be necessary for more specialized roles or later career advancements into more senior roles. Software developers must be analytical and creative to meet a business' needs properly. Software developers must be detail-oriented as they often work across multiple organizational applications to resolve errors.

2. **Needs**

    1. Once again, I already meet the degree requirements as I have a bachelor's degree in computer information systems. However, obtaining a master's in data analytics will help me stand out from other candidates and demonstrate my specialization in data science. This will allow me to easily apply for more senior-level roles and later move into leadership positions. The MSDA program will help me strengthen my analytical and creative thinking skills and improve my detail orientation through repeated application while completing course tasks.

Potential Career Goals Based on Strengths and MSDA Data Engineering

C. Identify a potential career goal in your career plan based on your strengths and academic/MSDA track interests.

My career goal is to become a data engineer. I believe the MSDA program's Data Engineer specialization track will give me the best return on the invested time and money. A master's degree with a data engineering specialization shows future employers my commitment to the role. It helps recruiters overlook a lack of experience in the actual role of a data engineer. The data engineering track will also help me build knowledge around relevant tools such as SQL, Python, R, and Tableau. It will also equip me with practical skills in creating and managing data processing (ETL pipelines), working with cloud databases, and learning the best practices for analyzing data on a large scale.

C1. Reflect on your career strengths as identified in your personalized CliftonStrengths assessment results.

Based upon my CliftonStrengths assessment, my top **five** strengths are **Relator**, **Responsibility**, **Significance**, **Restorative**, and **Belief** (CliftonStrengths, 2025). These strengths align with my career goal of becoming a data engineer and will support my professional development.

1. As a **Relator**, I thrive in environments where I can accomplish things with those with whom I have built strong relationships. Data engineers often work with cross-functional teams and make connections across organizations over their careers. While these relationships can be invaluable in multiple ways, in the office setting, strong relationships can lead to smoother collaboration, access to key resources, and opportunities to work on high-visibility projects.
2. My strength in **Responsibility** means I take ownership of tasks I commit to and feel bound to follow through on commitments. This is a great talent as a data engineer, as

no employer will want a half-finished data pipeline designed and implemented. This also builds on top of being a Relator above; when someone asks me to take on projects, they know I will go above and beyond to complete the project on time and to a standard we will both be satisfied with.

3. My desire for **Significance** motivates me to do my best and be recognized as a knowledgeable contributor. This drive is why I am pursuing my master's degree. It drives me to stay on top of the latest technology and methods of data analytics as well as the internal business goals of my organization.
4. Having strong **Restorative** talents means I thrive in solving problems and do not shy away when another issue arises. I enjoy troubleshooting and thinking of creative solutions to fix issues. Even if my ideas fail initially, I am one step closer to the solution. A data engineer must enjoy troubleshooting and implementing different fixes to resolve problems of varying priority levels.
5. Finally, my strong **Belief** talent means I have an enduring set of principles that I live by and will not easily waiver from these beliefs. Strong beliefs are foundational in relationships, allowing others to know what they can expect from me and that I am dependable and trustworthy. A data engineer will have access to millions of rows of private data and be responsible for the infrastructure supporting it. Organizations must know that their data engineers are technically capable, ethically sound, and reliable.

References

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